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Tuwaiq Academy

DATA SCIENCE & ARTIFICIAL INTELLIGENCE BOOTCAMP

AI Changed the Game.
But What's the Real Story?

Exploring AI Adoption, Productivity, and Trust Among Developers

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BOOTCAMP	Data Science & Artificial Intelligence Bootcamp
PROJECT	Final Project – Unit 3: Data Exploration & Dashboard Development
DATASET	Stack Overflow Developer Survey 2024
DATE	October 2026

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1. Introduction

1.1 Dataset

Table 1. Dataset summary.

Item	Details
Dataset title	Stack Overflow Developer Survey 2024
Source	https://survey.stackoverflow.co/2024/
File used	survey_results_public.csv (152.14 MB)
Size	65,437 respondents (rows) and 114 columns

The Stack Overflow Developer Survey is an annual survey of people who write code. Each row is one respondent. The 2024 survey includes questions about work, background and tools, and a group of questions about AI tools: whether people use them, which benefits and challenges they see, and how much they trust the accuracy of AI output.

Why I chose it. AI tools are now part of many developers' daily work, and this survey lets me look at AI adoption, productivity and trust in one real dataset. It is large enough to compare groups, and it has both numeric and categorical variables, which suits a full data exploration and dashboard project.

1.2 Problem definition

What is the dataset about? It is a self-reported survey of 65,437 developers. For this project I focus on ten columns about AI use, developer role, country, professional experience and salary.

Main questions.

- How many developers use AI tools?
- Does AI adoption differ by experience level and by developer role?
- Which benefits and which challenges do developers report?
- How much do AI users trust the accuracy of AI output, and which roles distrust it most?
- Are AI use and salary related in the United States, and how does experience relate to salary?

Insights I expected. I expected AI use to be common but not universal, productivity to be a leading benefit while trust is a leading concern, and salary to relate more to experience than to AI use. The analysis tests these expectations; it does not assume them.

1.3 Tools

The analysis was done in a Jupyter Notebook with Python (pandas, NumPy, Matplotlib and Seaborn). The interactive dashboard was built with Streamlit and Plotly.

2. Data Exploration

2.1 Structure of the raw data

The raw file has **65,437 rows and 114 columns** (*df.shape*). The data types are 100 text columns, 13 decimal-number columns and 1 integer column (*df.info()*). The notebook found **0 duplicate rows** in the raw data.

The table below shows the first five rows of the ten columns used in this project. These rows are the same in the raw and cleaned files, because no rows were removed. Long answers are shortened.

Table 2. First five rows (five of the ten selected columns). – means the answer is missing.

#	AISelect	DevType	YearsCodePro	Country	AIAcc
0	Yes	–	–	United States of Amer…	–
1	No, and I don't plan …	Developer, full-stack	17	United Kingdom of Gre…	–
2	No, and I don't plan …	Developer Experience	27	United Kingdom of Gre…	–
3	Yes	Developer, full-stack	–	Canada	Somewhat trust
4	No, and I don't plan …	Developer, full-stack	–	Norway	–

2.2 Summary statistics

The raw file has many numeric columns, mostly survey scores. For this project two numeric variables matter: years of professional coding (after conversion to numbers, see Section 3) and yearly salary in US dollars.

Table 3. Summary statistics from *describe()*, before salary outliers are removed.

Statistic	YearsCodePro_clean (years)	ConvertedCompYearly (USD)
Valid values (count)	51,610	23,435
Mean	10.21	86,155
Standard deviation	9.08	186,757
Minimum	0.5	1
25th percentile	3	32,712
Median	7	65,000
75th percentile	15	107,972
Maximum	51	16,256,600

> The salary mean (about 86,155 USD) is far above the median (65,000 USD), and the maximum is over 16 million USD. A few extreme salaries pull the mean up. This is why salary outliers are handled before any salary analysis (Section 3).

2.3 Missing values

Many columns in the raw data have a lot of missing values, because most survey questions are optional or shown only to some respondents. The columns with the most missing values include several about AI integration (for example, 98.25% missing for one of the AINext columns) and **ConvertedCompYearly** (64.19%).

Table 4. Missing values in the ten selected columns (65,437 rows).

Column (selected for this project)	Missing count	Missing %
ConvertedCompYearly (yearly salary, USD)	42,002	64.19
AIBen (AI benefits)	28,543	43.62
AIAcc (trust in AI accuracy)	28,135	43.00
AIChallenges (AI challenges)	27,906	42.65
AIThreat (AI as a job threat)	20,748	31.71
AISent (sentiment toward AI)	19,564	29.90
YearsCodePro (years of professional coding)	13,827	21.13
Country	6,507	9.94
DevType (developer role)	5,992	9.16
AISelect (AI adoption)	4,530	6.92

> Missing values are not random gaps only. The questions about AI accuracy and AI benefits were answered only by people who currently use AI tools (in the cleaned data, every one of these answers comes from an AI user). The AI challenges and AI threat questions were also answered by some people who plan to use AI soon. So each analysis uses only the people who were asked and answered that question.

3. Data Cleaning

The cleaning followed the notebook step by step. The goal was to keep the data that answers the research questions and to avoid deleting information that is still useful.

Table 5. Cleaning steps and their results.

Step	What was done	Result
1. Select columns	Kept the 10 columns needed for AI adoption, trust, benefits, challenges, role, experience, country and salary.	114 → 10 columns; 65,437 rows kept
2. Missing values	Checked the missing values. They were **kept** and handled separately inside each analysis, so each chart uses only valid answers. No values were filled in or invented.	Missing counts unchanged (Section 2.3)
3. Duplicates	The raw data has 0 duplicate rows. On the 10 selected columns alone, 9,630 rows look identical. They were kept, because the full raw rows are not duplicates (the rows differ in columns that were not selected).	No rows removed
4. Convert data type	YearsCodePro was stored as text. "Less than 1 year" became 0.5 and "More than 50 years" became 51. The rest were converted to numbers.	New column YearsCodePro_clean: 51,610 valid values
5. Create experience level	Grouped years into four levels: Beginner (under 3 years), Intermediate (3 to under 10), Experienced (10 to under 20) and Veteran (20 or more).	Intermediate 20,236; Experienced 13,327; Beginner 9,663; Veteran 8,384; missing 13,827
6. Salary outliers	Used the IQR rule on ConvertedCompYearly: upper limit = Q3 + 1.5 × IQR = 220,860.75 USD (the lower limit is negative, so it removes nothing). 978 salaries above the limit were excluded **from the salary analysis only**.	Salary dataset: 22,457 rows
7. Save the cleaned file	Saved the project dataset as cleaned_data.csv.	65,437 rows × 12 columns (10 original + 2 new)

Steps not applied. The notebook did not impute (fill in) missing values, did not remove rows with missing values, and did not rename columns; the original column names were kept. Missing values were therefore examined and kept, then handled separately in each analysis.

> Before cleaning: 65,437 × 114 (raw). After cleaning: 65,437 × 12 (all rows kept; 10 selected columns plus 2 new variables, YearsCodePro_clean and ExperienceLevel). The missing-value counts of the selected columns are the same before and after.

4. Exploratory Data Analysis (EDA)

All figures in this section come from my notebook. Percentages use only respondents who answered the question in each chart (the valid responses), not the full 65,437 rows. The sample size is given when it matters.

4.1 Univariate analysis

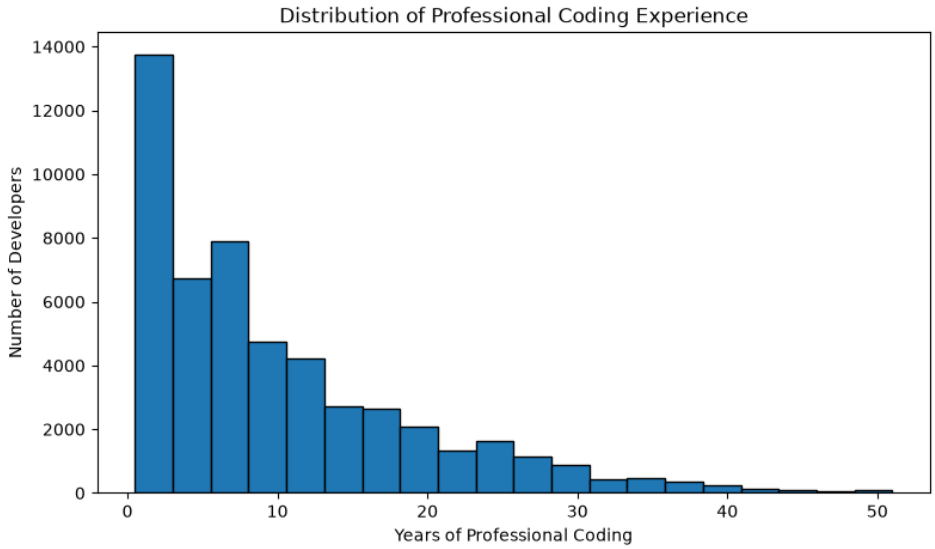


Figure 1. Distribution of professional coding experience (years).

Most developers have few years of professional coding experience. The distribution is right-skewed: many people have 0 to 10 years, and fewer have 30 years or more. The median is 7 years (51,610 valid answers).

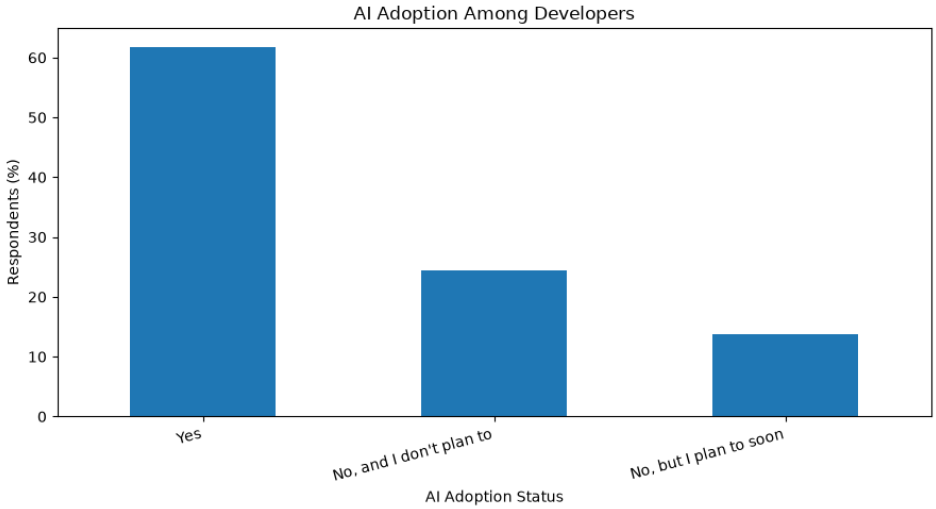


Figure 2. AI adoption among developers.

61.8% of respondents use AI tools (37,662), 24.4% do not plan to (14,837) and 13.8% plan to start soon (8,408). Percentages are based on the 60,907 respondents who answered this question.

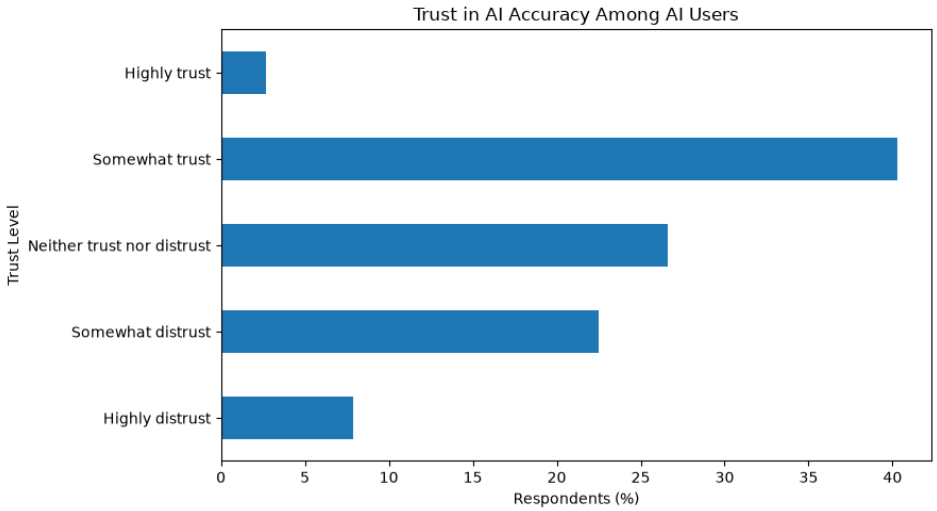


Figure 3. Trust in the accuracy of AI output among AI users.

Among the 37,302 AI users who answered, 43.0% trust AI accuracy somewhat or highly (40.3% somewhat, 2.7% highly) and 30.4% distrust it (22.5% somewhat, 7.9% highly). Only 2.7% highly trust it, and 26.6% are neutral. Trust is cautious.

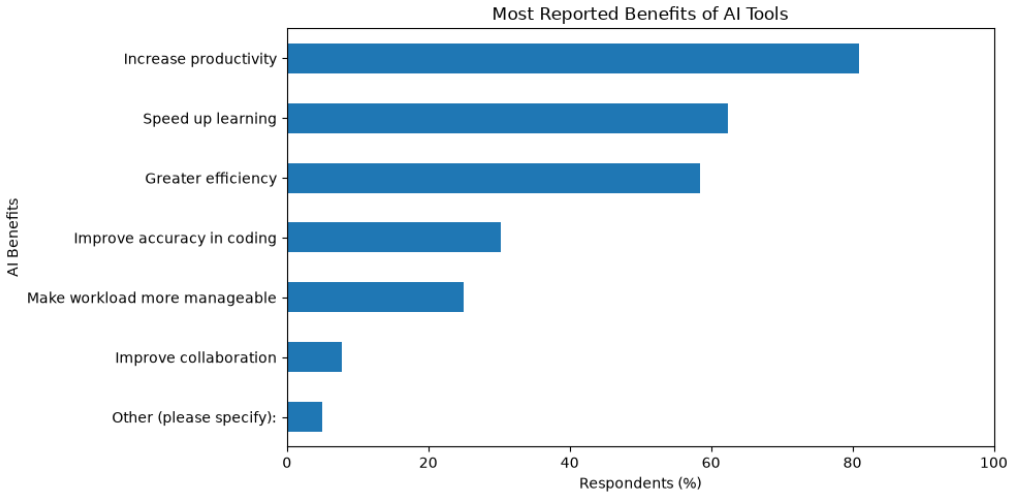


Figure 4. Most reported benefits of AI tools (multiple answers allowed).

Increase productivity is the most reported benefit (81.0%), followed by speeding up learning (62.4%) and greater efficiency (58.5%). Improving accuracy in coding was selected by 30.3%. Percentages are of the 36,894 AI users who answered, and add up to more than 100% because people could select several benefits. These are benefits reported by developers, not measured improvements.



Figure 5. Most reported challenges of AI tools (multiple answers allowed).

Not trusting the output (65.1%) and a lack of context about the codebase (61.9%) are the two most reported challenges. Missing security policies (32.5%) and a lack of training (31.6%) follow. The base is the 37,531 respondents who answered the challenges question, and benefits and challenges are separate survey questions.

4.2 Bivariate analysis

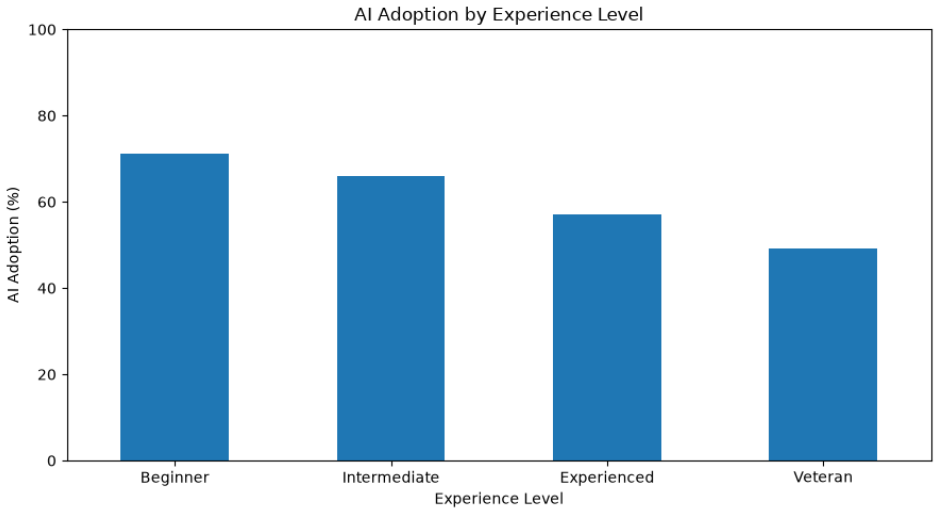


Figure 6. AI adoption by experience level.

AI adoption falls at every step up in experience: 71.2% for Beginners, 66.0% for Intermediate, 57.2% for Experienced and 49.2% for Veterans, a gap of 22 percentage points. This is an association in the data. It does not show that experience causes lower AI use.

Table 6. AI adoption by developer role, for the 10 most common roles (notebook output).

Developer role (10 most common)	Respondents in role	Use AI (%)
Developer, front-end	3,349	69.1
Developer, full-stack	18,260	65.3
Developer, mobile	2,021	65.2
Engineering manager	1,275	62.5
Developer, back-end	9,928	62.0
Student	5,102	61.5
Academic researcher	1,238	57.8
Other (please specify)	2,458	51.3
Developer, desktop or enterprise applications	2,493	45.0
Developer, embedded applications or devices	1,623	43.5

Front-end developers have the highest adoption (69.1%) and embedded developers the lowest (43.5%) among the ten most common roles. The percentage is the share of people in each role who answered "Yes" to using AI, among those who answered the question.

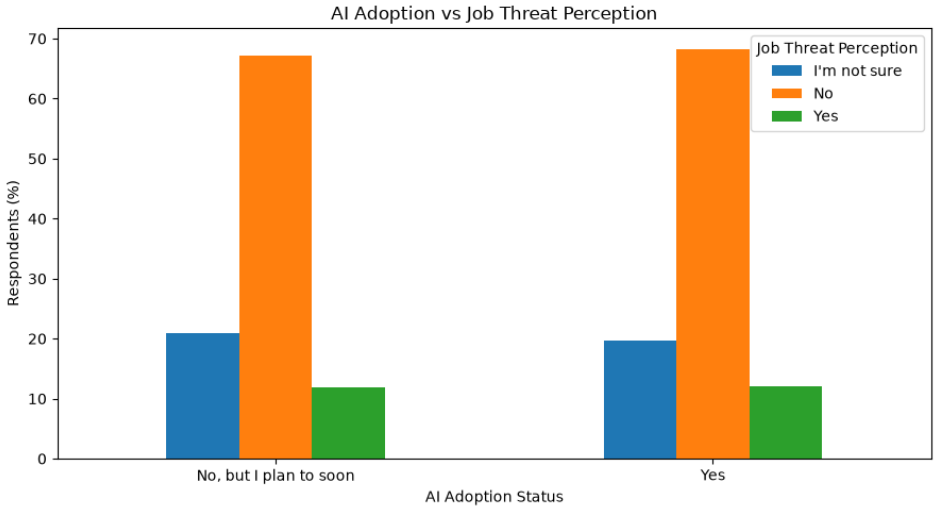


Figure 7. AI adoption status and perceived job threat from AI.

About 68% of AI users do not see AI as a threat to their job (68.3%), 12.1% do, and 19.6% are not sure. The pattern is similar among people who plan to use AI soon (67.1%, 11.9%, 21.0%). The question was not answered by people who do not plan to use AI.

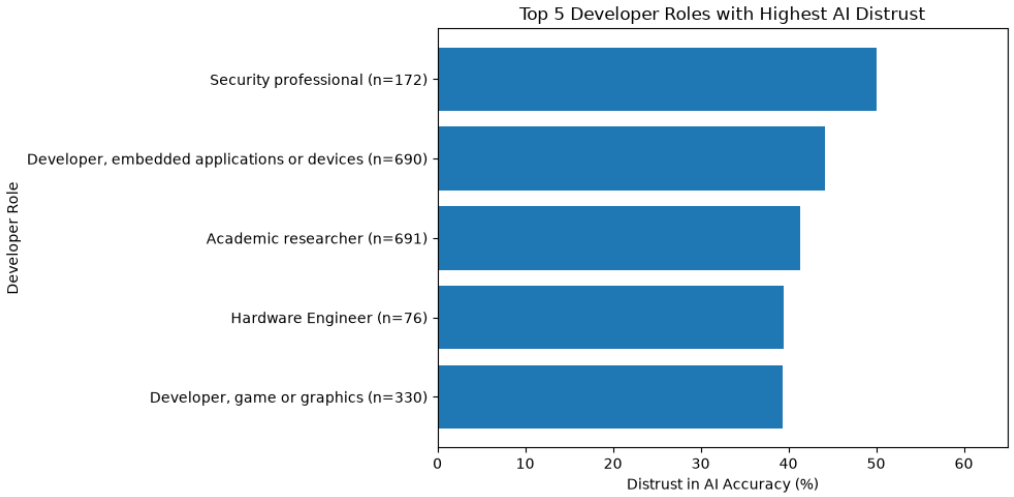


Figure 8. The five developer roles with the highest distrust of AI accuracy.

> Security professionals show the highest distrust (50.0%, n = 172), followed by embedded developers (44.2%, n = 690), academic researchers (41.3%, n = 691), hardware engineers (39.5%, n = 76) and game or graphics developers (39.4%, n = 330). Distrust means "somewhat" or "highly" distrust. Groups with small samples, such as hardware engineers, should be read with care.

Salary analysis

Salary analysis uses the salary dataset without outliers (22,457 rows). Across all countries, the median salary was lower for AI users than for non-users. Because salaries differ a lot between countries, the comparison was repeated inside the United States, the country with the most salary records (3,990).

Table 7. Median yearly salary by AI adoption group (salary outliers excluded).

AI adoption group	All countries: n	All countries: median (USD)	US: n	US: median (USD)
Uses AI	14,225	59,073	2,266	130,000
Plans to use AI soon	2,923	63,370	501	131,000
Does not plan to use AI	5,309	72,001	1,223	128,731

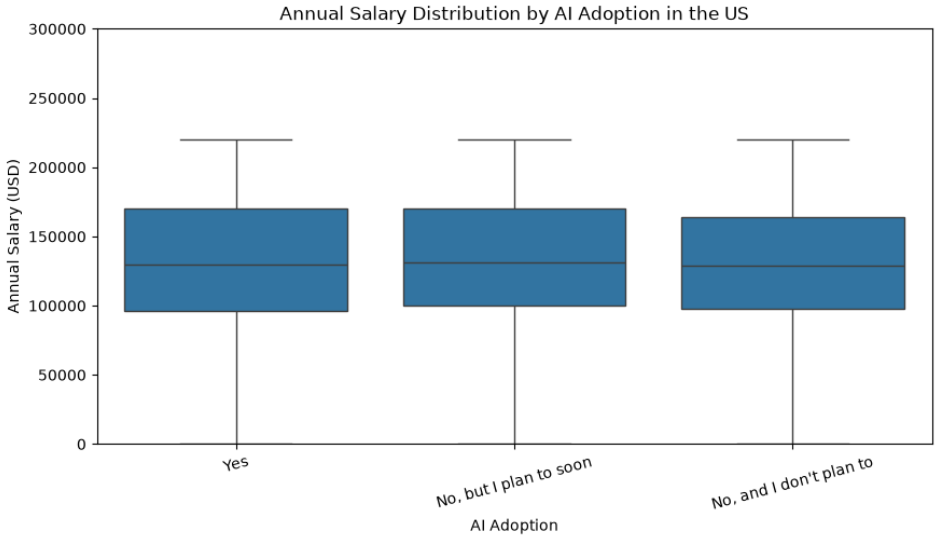


Figure 9. Annual salary distribution by AI adoption in the United States.

In the United States, the salary distributions of the three groups are very similar, with medians close to 130,000 USD. The lower all-countries median for AI users may partly reflect differences in the countries of respondents, so it should not be read as an effect of AI. These differences do not show that AI causes higher or lower salaries.



Figure 10. Professional coding experience and annual salary in the United States.

Salaries tend to rise with years of professional experience, but they vary a lot between people with similar experience.

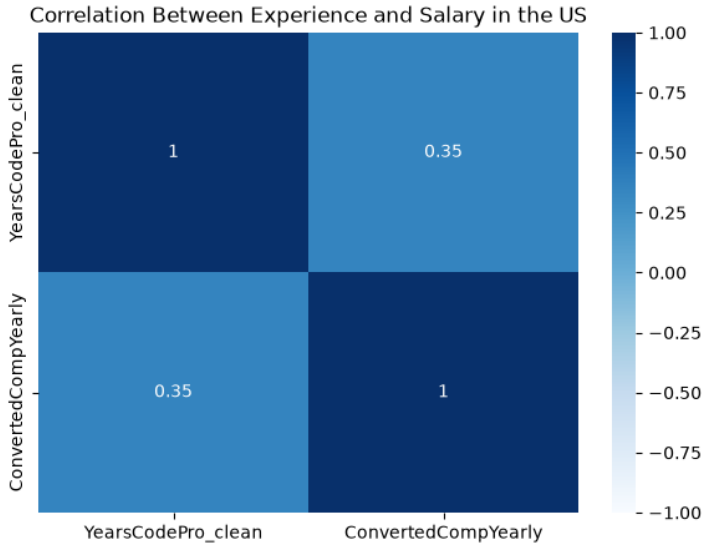


Figure 11. Correlation between experience and salary in the United States.

The correlation is r = 0.35, a moderate positive relationship. Salary tends to increase with experience, but experience does not explain salary differences on its own.

4.3 Multivariate analysis

The multivariate analysis looks at three variables together in the United States: developer role, experience level and AI adoption. Only groups with at least 20 AI users and 20 non-users were compared, which left ten groups.

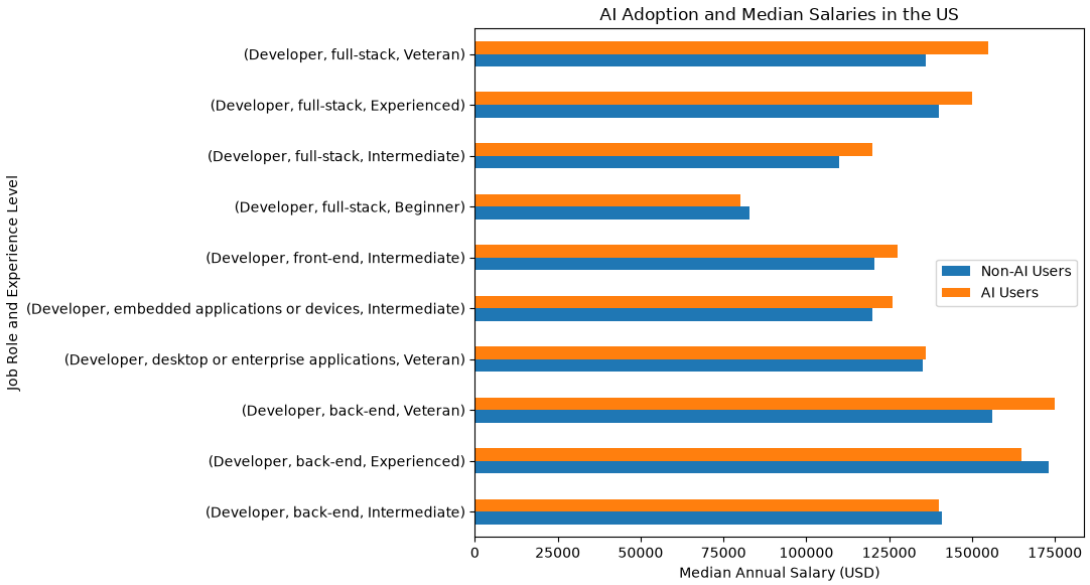


Figure 12. Median salary of AI users and non-users by developer role and experience level (United States).

AI users had a higher median salary in seven of the ten comparable groups; in the other three they did not. AI adoption is therefore not consistently linked to lower salaries, but the differences vary by role and experience. This is an observed pattern. It does not mean that using AI causes higher salaries.

5. Dashboard Overview

The Streamlit dashboard turns the notebook analysis into an interactive story. Visitors can follow the analysis in order, or explore a group of developers that interests them.

Live dashboard: https://ai-developer-dashboard-5sritqcuaglx8appvgfzlmr.streamlit.app/
GitHub repository: https://github.com/GHADI-AI26/AI-Developer-Dashboard

5.1 Purpose and layout

The dashboard opens with a hero section that states the project question, followed by a navigation bar. The main page is organized in numbered sections, each starting with a question and ending with a short insight and a note on why it matters:

- **Overview and Explore:** summary statistics, data preview and the "Explore Your Developer World" tool.
- **Data preparation:** a before-and-after view of the cleaning steps.
- **AI adoption, experience and roles:** the adoption analyses from the notebook.
- **Benefits and challenges, productivity and trust, accuracy, cybersecurity trust gap and job threat.**
- **Salary:** bivariate and multivariate salary comparisons.
- **Find Yourself in the Data** and **The Trust Test** (interactive activities), and a 30-second summary.

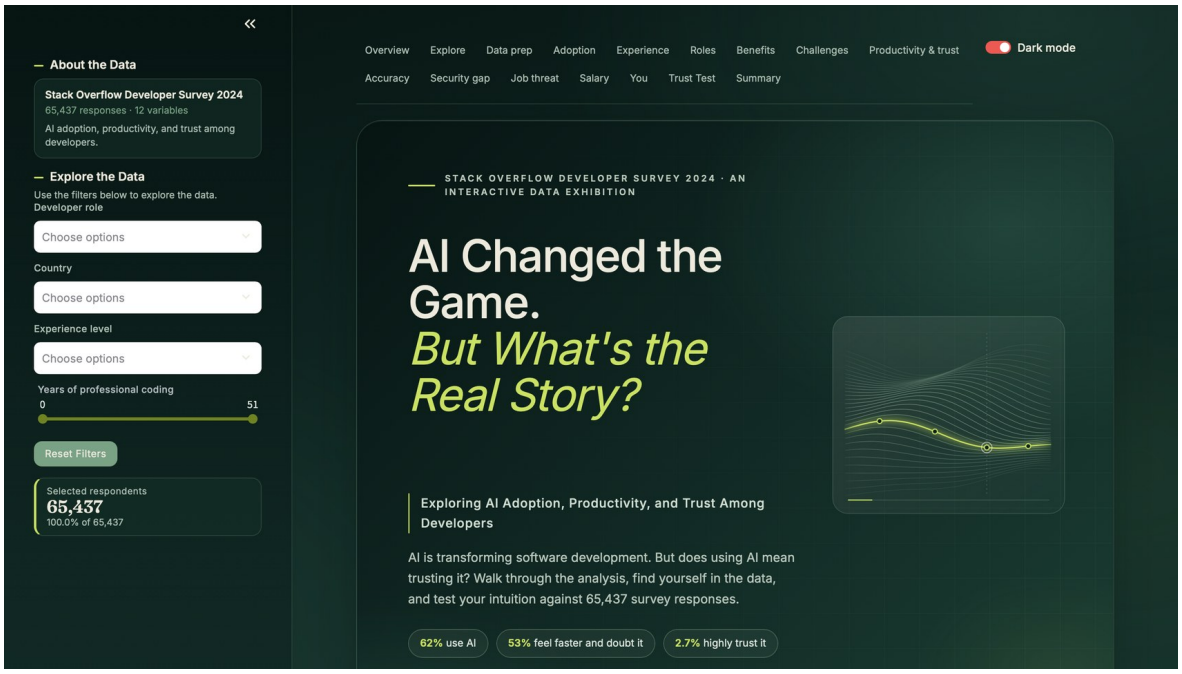


Figure 13. The dashboard landing view: hero section, navigation bar and the sidebar (dark mode, shown with no filter applied).

> The hero states the title and key figures. The navigation links jump to each section, and a toggle switches between dark and light mode.

5.2 Sidebar and filters

The sidebar describes the dataset and offers the filters required for the project. It contains a dataset description, a **multiselect** for developer role, country and experience level, and a **slider** for years of professional coding. A Reset Filters button clears all filters, and a counter shows how many respondents are selected. Selecting all options of a filter is treated the same as leaving it empty, so respondents with missing values are not dropped by accident. The filters update the main analyses; the Find Yourself, Trust Test and summary sections always use the full survey so their comparison stays stable.

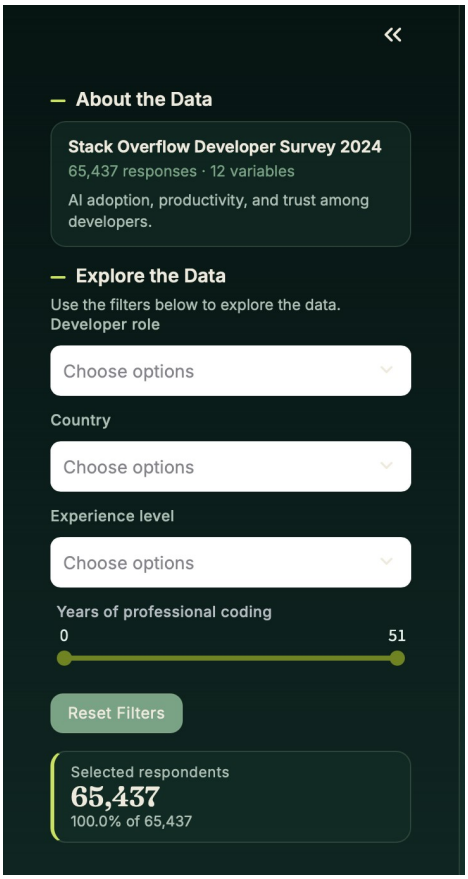


Figure 14. The sidebar: dataset description, filters and the respondent counter (cropped from Figure 13).

5.3 Interactive charts and statistics

The charts are interactive Plotly charts: hovering shows exact values and sample sizes. The main page also shows summary statistics and an expandable data preview. Small groups are handled carefully: a comparison is shown only when it has at least 30 valid answers (20 for salary groups, as in the notebook).

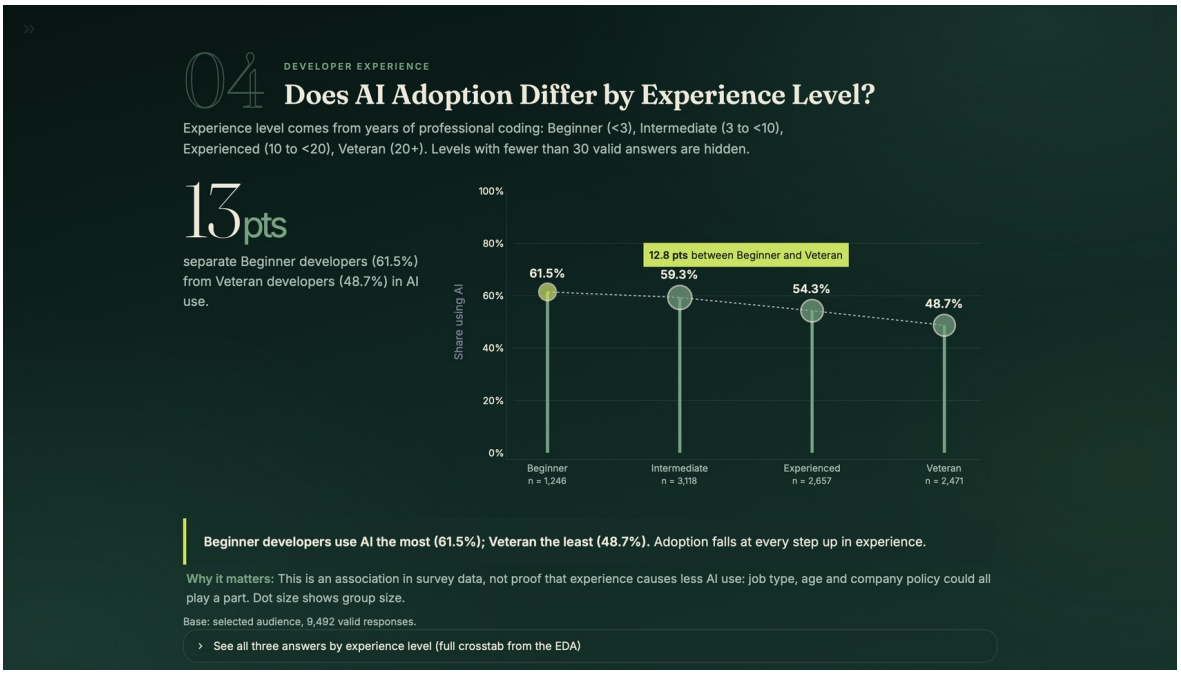


Figure 15. The AI adoption by experience level section, with the sidebar filtered to the United States.

> The section reproduces the notebook finding for the filtered group: adoption falls from Beginner to Veteran. The chart, the large number, the insight and the base size all update with the filter.

5.4 Exploring different developer groups

The **Explore Your Developer World** section lets visitors choose a country, a developer role and an experience level from searchable dropdowns. It shows the number of respondents, the AI adoption rate, trust, and reported benefits and challenges for that group, and compares each result with the full survey. The full-survey baseline never changes, and groups with fewer than 30 valid answers are not shown as percentages.



Figure 16. The Explore Your Developer World section (here showing all countries and all roles).

> With no selection, the group equals the full survey, so every difference is 0.0 points: 65,437 respondents, 61.8% use AI, 43.0% trust AI accuracy, and 81.0% report a productivity benefit (the same values as in the notebook).

5.5 How the dashboard supports the research questions

The dashboard also adds one analysis that is not in the notebook: how often respondents report a productivity benefit and also report not trusting the output. It uses the 30,327 respondents who answered both the benefits and the challenges questions. Of these, 16,084 (53.0%) selected both "Increase productivity" and "Don't trust the output or answers". Among the 24,703 who selected productivity, 65.1% also selected distrust. This describes how answers overlap; it does not show cause and effect.



Figure 17. The Productivity and Trust section: two circles drawn to scale, with the sidebar filtered to the United States (5,163 respondents).

> The overlap of the circles shows respondents who report productivity and also distrust the output. The numbers in this screenshot are for the US group, so they differ from the full-survey values above.

Table 8. Research questions and the dashboard sections that answer them.

Research question	Where it is answered in the dashboard
How many developers use AI?	AI adoption section; Explore section; sidebar counter
Does adoption differ by experience and role?	Experience and Roles sections; filters
Which benefits and challenges are reported?	Benefits and Challenges sections; Explore section
How much do developers trust AI accuracy?	Accuracy and trust, Productivity and trust, Cybersecurity trust gap
Is AI use related to salary?	Salary section (all countries, United States, and multivariate)

6. Insights & Conclusion

6.1 Main findings

- **AI use is common but not universal.** 61.8% of respondents who answered use AI tools; 13.8% plan to start soon.
- **Adoption is lower with more experience.** It falls from 71.2% (Beginner) to 49.2% (Veteran). Roles also differ: from 69.1% (front-end) to 43.5% (embedded) among the ten most common roles.
- **Productivity is the main reported benefit; trust is the main concern.** 81.0% report increased productivity, while 65.1% name not trusting the output as a challenge.
- **Trust is cautious.** Only 2.7% of AI users highly trust AI accuracy; 43.0% trust it somewhat or highly and 30.4% distrust it. Security professionals show the highest distrust (50.0%, n = 172).
- **Salary patterns are not explained by AI use.** In the United States the median salary is about 130,000 USD in all three AI adoption groups. Salary is moderately related to experience (r = 0.35).
- **Outliers.** A few extreme salaries (978 above 220,860.75 USD) distort the mean, so salary comparisons use medians and exclude these outliers.

6.2 Limitations

- The data is a self-selected survey. It may not represent all developers, and answers are self-reported.
- Many questions have missing answers, and some questions were asked only of AI users or of people planning to use AI. Percentages therefore describe the people who answered, not all respondents.
- Benefits and trust are opinions, not measured results. A reported productivity gain is not proof of a real gain.
- Salary has 64.19% missing values and extreme values. Salary comparisons use only the 22,457 records without outliers, and salaries in different countries are not directly comparable.
- The analysis shows associations only. It cannot show that AI use causes any difference in trust, productivity or salary.
- The project uses 10 of the 114 columns, so other factors (age, education, company size) are not considered. The 9,630 duplicate-looking rows in the selected columns were kept.

6.3 What I learned

- How to check data structure and missing values before analysis, and why missing values should be understood before they are removed or filled.
- How to use the right denominator: each question must be calculated only on the people who answered it.
- How to compare groups carefully, and why small groups and outliers can give misleading results.
- How to turn a notebook analysis into an interactive Streamlit dashboard with filters, charts and insights.

6.4 Recommendations for future analysis

- Compare the 2024 results with earlier survey years to see how AI adoption and trust change over time.
- Add more variables, such as age, education and company size, and analyse them together (for example with a regression model).
- Study trust in more detail by tool type and by how often AI is used.

- Test whether differences between groups are statistically significant, and extend the analysis to countries other than the United States.

6.5 Conclusion

AI tools are widely used by developers and are valued mainly for productivity, but trust in their output is cautious, especially among more experienced developers and some roles such as security professionals. In the United States, salary is more clearly related to experience than to AI use. These findings describe patterns in survey answers; they do not prove cause and effect. The dashboard lets anyone explore these patterns for the groups that interest them.

7. References

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